



# Characteristics of soil preferential flow and its influencing factors under different elevation gradients in Mopan Mountain, Central Yunnan, China

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## Abstract

**Aims** This study aims to investigate the morphological characteristics of soil preferential flow and its influencing factors under different elevation gradients in Mopan Mountain, central Yunnan.

**Methods** Three elevation gradients—high (2565 m), medium (2011 m), and low (1630 m)—were selected as research plots. Dominant native vegetation communities were identified at each site, and field dye tracer experiments were conducted to assess soil preferential flow. Based on multiple field surveys, the dominant vegetation types were classified as alpine meadows at high elevations, evergreen broad-leaved forests at mid elevations, and mixed tree–shrub forests at low elevations. Key research indicators included quantitative parameters of preferential flow,

basic soil physicochemical properties, and standard root system metrics.

**Results** Preferential flow was observed across all elevation levels, with the most pronounced development occurring at mid elevation, where the stained area ratio reached 37.98% and the preferential flow ratio was 48.84%. The low-elevation plot exhibited moderate preferential flow development, while the high-elevation meadow showed the weakest. Among the four root indicators, root volume density contributed most significantly (82.1%). Spatial differentiation in soil physicochemical properties across elevations.

**Conclusion** Soil structural variation driven by elevation gradients is a key factor controlling the spatial heterogeneity of preferential flow. However, the influence of plant roots is also critical. Root-created channels serve as important preferential pathways that significantly enhance soil water.

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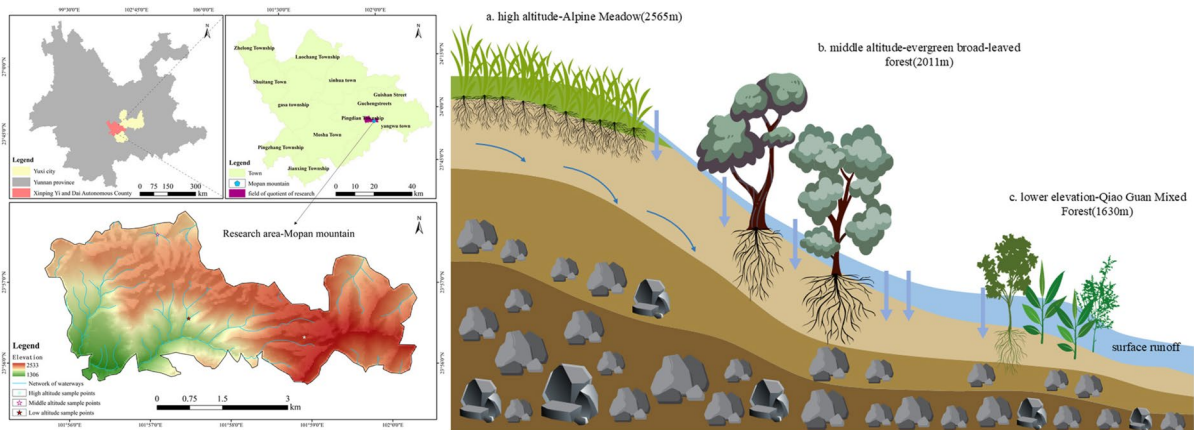
## Introduction

The phenomenon of preferential flow in soil water movement refers to the rapid migration of fluids through dominant pathways—such as macropores, fractures, or biopores—driven by gravity or pressure gradients, bypassing the surrounding soil matrix.

This concept was initially introduced by Hill (Hill and Parlange 1972) in the 1960s and was later systematically developed by Beven and Germann (Beven and Germann 2013), forming a foundational paradigm in hydrological research. The theoretical development of preferential flow has evolved through three main stages: early morphological characterization using dye tracers (Flury and Flühler 1995), intermediate quantification of pore networks through CT scanning techniques, and recent mechanistic investigations integrating isotope tracing with controlled laboratory simulations (Sprenger et al. 2015). Over decades of advancement, the study of preferential flow has expanded from simple hydrological observations to a multidisciplinary domain encompassing soil science, hydrology, and environmental science. The occurrence of preferential flow is influenced by a range of biotic and abiotic factors. Biologically, root architecture is a key regulator of soil preferential flow (Mair et al. 2022; Wu et al. 1999). Stable macro pore networks are often formed when plant root diameters exceed 0.5 mm (Bundt et al. 2001), while soil fauna such as termites create additional preferential channels through their burrowing activity (Jouquet et al. 2016). Physically, soils with sand content exceeding 60% are prone to finger flow formation, whereas clay content above 15% significantly inhibits water movement (Chen et al. 2021). Chemically, high levels of organic matter may reduce pore connectivity due to cementation effects; reductions in organic matter are often associated with decreased saturated hydraulic conductivity (Bargués Tobella et al. 2014). Environmental factors also play a key role. Dormant pores may become activated when rainfall intensity surpasses a critical threshold (Hardie et al. 2013), and repeated freeze–thaw cycles increase crack density, further enhancing preferential pathways (Leuther and Schlüter 2021). From a geomorphological perspective, preferential flow significantly alters slope erosion dynamics by modifying water transport routes. When rainfall intensity exceeds the soil's infiltration capacity, water rapidly travels through preferential pathways, intensifying subsurface runoff and triggering piping erosion, which can destabilize the soil structure (Bernatek-Jakiel and Poesen 2018). These interacting factors collectively influence the formation and persistence of preferential flow, ultimately shaping soil hydrology and ecosystem processes. For instance, macropore flow—a common type of

preferential flow—has a profound impact on subsurface sediment transport. Valentin et al. reported that in areas with well-developed macropore networks, subsurface sediment transport can be up to 3.8 times greater than in matrix flow conditions (Valentin et al. 2005). Gyssels et al. further demonstrated that preferential flow not only accelerates clay leaching but also weakens the root anchoring system within the soil. The structural consequences are especially pronounced in steep terrain (Gyssels et al. 2005). Zhu et al. found that in steep slopes of Mopan Mountain, Central Yunnan, preferential flow significantly reduces the shear strength of surface soils, thereby increasing the likelihood of shallow landslides (Zhu et al. 2023). Given these wide-ranging impacts, it is essential to establish a comprehensive and systematic theoretical framework for understanding soil preferential flow.

In recent years, increasing attention has been directed toward understanding how altitudinal gradients regulate root growth and development as well as soil preferential flow characteristics, particularly within mountain ecosystems and soil hydrology research. Studies have demonstrated that along elevational gradients, significant differences occur in soil physicochemical properties at different altitudes (Chang et al. 2025). Moreover, He et al. (He et al. 2025) reported that the  $\alpha$ -diversity of soil bacterial and fungal communities exhibits a unimodal pattern in response to increasing elevation, thereby exerting a substantial influence on soil physicochemical properties and microbial community structures in the study area. In parallel, substantial progress has been made in preferential flow research across diverse regions and ecosystems. For example, Zhang et al. found that in desert oases, soil structural properties and moisture content jointly determine preferential flow under alternating dry and wet conditions (Zhang et al. 2017). In a subtropical monsoon region, Deng et al. investigated fir plantations of varying stand ages and discovered that the degree of preferential flow development decreases with increasing forest age, highlighting the dominant role of lateral flow in groundwater recharge within fir forests (Deng et al. 2025). Despite these advances, previous studies have largely focused on the effects of a single altitudinal gradient on preferential flow or examined preferential flow across different soil types and regions. Systematic investigations into the coupled variations of



**Fig. 1** Overview of the study area and three-dimensional schematic diagram. (a) represents the high-altitude herb area, (b) represents the middle altitude evergreen broad-leaved forest area, (c) represents the low altitude mixed forest area

vegetation root development, soil factors, and preferential flow characteristics along multiple altitudinal gradients remain limited. Furthermore, the quantitative effects of elevation-driven changes—mediated through vegetation root systems and soil structure—on the development processes of preferential flow have yet to be clearly elucidated. These knowledge gaps urgently call for comprehensive research through integrated field sampling, in situ observations, laboratory experiments, and numerical modeling.

Most existing studies have primarily focused on preferential flow phenomena within single research areas or specific elevation ranges, or have been limited to examining the mechanisms of individual influencing factors. However, the understanding of coupled interactions between root systems and preferential flow development across different elevation gradients and soil textures remains incomplete. Elevation integrates hydrothermal conditions, vegetation distribution, and soil characteristics, serving as a key regulatory factor for preferential flow, yet its underlying mechanisms are still insufficiently explored. This knowledge gap is particularly pronounced in subtropical mountainous regions such as Mopanshan Mountain in central Yunnan, where pronounced vertical zonation exists. The interlinked processes among altitudinal gradients, root architecture, soil properties, and preferential flow variation—as well as the dynamic mechanisms of preferential flow under the coupling effects of multiple factors—have not yet been systematically investigated. Therefore, it is

crucial to explore the morphological characteristics of preferential flow and its influencing factors across altitudinal vegetation zones by integrating dye tracer techniques with root scanning analyses. Such an integrated approach will not only advance the theoretical framework of soil hydrology in mountainous ecosystems but also address the unresolved scientific question of how the synergistic interactions between root systems and soil structure govern the evolution of preferential flow along altitudinal gradients. Moreover, the findings will provide essential parameters for hydrological process modeling in mountain environments and offer a theoretical foundation for hydrological regulation and ecological restoration in degraded regions.

## Materials and methods

### Overview of the study area

The study was conducted in Mopanshan National Forest Park, located in Xiping County, within the southeastern part of the Xiping Yi and Dai Autonomous County, Yuxi City, Yunnan Province, China. The area lies on the southern edge of the Yunnan Plateau, east of the Ailao Mountains and the Yuanjiang River Valley, as shown in Fig. 1. Geographically, it spans from 101°55′13″E to 102°4′E and from 23°54′N to 24°2′30″N, with elevations ranging from 1,260 m to 2,614 m. The region experiences a

subtropical plateau monsoon climate, characterized by year-round mild temperatures, distinct wet and dry seasons, and synchronization of rainfall and thermal peaks. The majority of precipitation occurs between May and September, with an average annual temperature ranging from 17 °C to 20 °C. Vegetation diversity varies significantly along the elevation gradient. Dominant plant species include *Pinus yunnanensis*, *Ageratina adenophora*, and *Lespedeza davidii*. Soils are primarily red earth, followed by yellow earth and brownish-yellow earth. The main forest types include evergreen broad-leaved forests, coniferous forests, and mixed conifer-broadleaf forests.

## Research Methodology

### Plot establishment

To reflect the variation in vegetation and soil properties across different elevations, three elevation bands were selected within the study area. The high-elevation plot (2,565 m) was dominated by herbaceous vegetation, the mid-elevation plot (2,011 m) by tree species, and the low-elevation plot (1,630 m) by mixed tree and shrub communities. In order to ensure that the elevation gradient has a large difference, the total elevation gradient between plots was approximately 1,000 m (Hu et al. 2023). At each elevation level, three replicate quadrats measuring 3 m×3 m were established, resulting in a total of nine quadrats. Replicates were spaced less than 5 m apart to minimize environmental variability. The low-elevation quadrats were designated D1, D2, and D3; the mid-elevation quadrats Z1, Z2, and Z3; and the high-elevation quadrats G1, G2, and G3. The basic information of the plots is shown in Table 1.

### Dye tracing experiment

Field investigations confirmed the feasibility of conducting dye tracing experiments at the study site, and dye application was scheduled during a period of no rainfall to avoid interference from precipitation. Prior to installing the iron frames, the surface of each sample plot was gently cleaned to remove litter such as dead branches and leaves, taking care not to disturb the natural structure of the topsoil. Methylene blue solution was used as the dye tracer. Two iron frames

(60 cm×60 cm×60 cm; wall thickness 0.5 cm) were embedded at different positions within each quadrat to obtain both vertical and horizontal dyeing profiles for the same plant type. The frames were inserted to a depth of 5–20 cm, and the soil within 5 cm of the inner wall was compacted using a wooden mallet to prevent lateral leakage of the dye solution along the frame edges, which could compromise the accuracy of the observations. A 4 g·L<sup>-1</sup> methylene blue solution (totalling 19 L per test) was applied evenly across the designated area within the iron frame using a Mariotte bottle to maintain a constant flow rate. The stained surface was then immediately covered with an insulating plastic film to prevent contamination and reduce evaporation. After 24 h, the plastic film was removed, and the iron frame was carefully dismantled. Soil profile excavation was then carried out. Due to the potential instability of soil moisture movement near the frame edges, which could lead to lateral seepage, the vertical profile was excavated within a 50 cm×50 cm central area inside the frame to ensure representative results. When excavating and photographing soil vertical profiles, the excavation was carried out at a horizontal width of 15 cm, with a total of 4 vertical profiles dug. After the excavation of each profile, it was trimmed with a small shovel. Each layer was photographed three times for image analysis. Excavation and image collection were terminated once no visible dye staining remained in the deeper layers.

### Determination of Preferential Flow Related Indicators

**Staining Area Ratio** The stained area ratio represents the percentage of the stained region in the vertical soil profile relative to the total profile area. This metric effectively reflects the overall dynamics of vertical movement of the dye solution through the soil. The processed images obtained from Photoshop were analyzed using Image-Pro Plus to extract stained area parameters. The stained and total areas were then calculated using the following formula. Formula:

$$DC = \left( \frac{D}{M} \right) \cdot 100\% \quad (1)$$

where *DC* is the stained area ratio (%); *D* is the stained area in the image; *M* represents the total area of the image.

**Table 1** Basic information of the sample plots

| Stand type                                              | Serial Number | Soil type         | Slope | Aspect of slope | Altitude(m) | Longitude(E) | Latitude(N) | Plant height(m) |
|---------------------------------------------------------|---------------|-------------------|-------|-----------------|-------------|--------------|-------------|-----------------|
| Herbaceous<br>( <i>Ageratina</i><br><i>Adenophora</i> ) | G1            | Brown-yellow soil | 17    | Southwest       | 2565        | 101.58.54    | 23.56.19    | 0.66 ± 0.08     |
|                                                         | G2            |                   | 17    | Southwest       | 2565        | 101.58.54    | 23.56.19    | 0.59 ± 0.05     |
|                                                         | G3            |                   | 17    | Southwest       | 2565        | 101.58.54    | 23.56.19    | 0.71 ± 0.06     |
| Arbor<br>(Evergreen<br>Broadleaf<br>Forest)             | Z1            |                   | 21    | Northeast       | 2011        | 101.57.05    | 23.57.35    | > 10            |
|                                                         | Z2            |                   | 21    | Northeast       | 2011        | 101.57.05    | 23.57.35    | > 10            |
|                                                         | Z3            |                   | 21    | Northeast       | 2011        | 101.57.05    | 23.57.35    | > 10            |
| Shrub<br>( <i>Lespedeza</i><br><i>davidii</i> )         | D1            | Red soil          | 26    | Southwest       | 1630        | 101.57.28    | 23.56.33    | 1.05 ± 0.11     |
|                                                         | D2            |                   | 26    | Southwest       | 1630        | 101.57.28    | 23.56.33    | 1.33 ± 0.07     |
|                                                         | D3            |                   | 26    | Southwest       | 1630        | 101.57.28    | 23.56.33    | 0.96 ± 0.14     |

**Preferential Flow Ratio** The preferential flow ratio refers to the proportion of the stained area attributed to preferential flow relative to the total stained area in the vertical soil profile. Based on bitmap analyses conducted in Image-Pro Plus, when the stained area ratio is  $\geq 80\%$ , it is defined as the matrix flow depth. The matrix flow depth parameter is subsequently substituted into the following formula to calculate the preferential flow ratio. Formula:(2).

$$PF_{fr} = \left(1 - \frac{UniFr \cdot W}{TotStAr}\right) \cdot 100\% \quad (2)$$

In the formula:  $PF_{fr}$  indicates the ratio of preferential flow;  $UniFr$  indicates the depth of matrix flow (cm) in the vertical profile image;  $W$  represents the width of the entire stained profile (cm), which is 50 cm in this study;  $TotStAr$  represents the total stained area ( $\text{cm}^2$ ) in the vertical cross-sectional image.

**Matrix Flow Depth** Matrix flow depth refers to the soil depth (cm) at which the matrix flow zone in the vertical soil profile first decreases to 80% of the maximum staining or lower, indicating the transition into the preferential flow zone.

**Average Maximum Infiltration Depth** This index represents the average value of the maximum infiltration depth reached by the dye in the vertical soil profile.

**Length index** The length index quantifies the heterogeneity of vertical dye distribution by summing

the absolute differences in stained area ratios between adjacent soil layers. The vertical soil profile is divided into equal segments. The vertical soil profile is divided into 500 layers, and the stained area ratio is calculated for each layer. The absolute difference between the  $n$ -th and  $(n+1)$ -th layers is then computed, and the sum of all absolute differences represents the length index. The index is calculated as follows:

$$Li = \sum_{i=0}^n |Dc(i+1) - Dci| \quad (3)$$

In the formula:  $Li$  represents the length index;  $n$  is the total number of vertical soil layers (e.g.,  $n = 500$  layers in this study, each 1 mm thick);  $Dc(i+1)$  and  $Dci$  represent the dye coverage ratio (%) of the  $(i+1)$  th layer and the  $(i)$ th layer of the dye profile, respectively.

### Sample Collection and Measurement

To minimize soil disturbance during the dye-tracing experiment, all soil sampling and measurements were conducted adjacent to the embedded iron frames. Sampling depth was determined based on the maximum dye penetration depth, with samples collected at 10 cm intervals down to 30 cm, resulting in three stratified soil layers. For each layer, approximately 500 g of mixed soil was collected using the five-point sampling method. Samples were air-dried, sieved, and stored for laboratory analysis of physicochemical properties. Plant root samples were collected separately and

**Table 2** The physicochemical properties of soils from each sample plot

| Plot              | Soil depth (cm) | Bulk density<br>(g•cm <sup>-3</sup> ) | Volumetric<br>water con-<br>tent (%) | MWD (%)                                 | Mechanical composition (%)              |                                                |                                               |
|-------------------|-----------------|---------------------------------------|--------------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------------|-----------------------------------------------|
|                   |                 |                                       |                                      |                                         | Clay                                    | Silt                                           | Sand                                          |
| High altitude     | 0–10            | 1.27 ± 0.06b                          | 18.44 ± 0.07b                        | 2.82 ± 0.25c                            | 17.96 ± 1.07a                           | 54.24 ± 1.99a                                  | 27.80 ± 3.00b                                 |
|                   | 10–20           | 1.32 ± 0.04b                          | 20.33 ± 0.07c                        | 3.19 ± 0.15b                            | 19.44 ± 2.32a                           | 54.52 ± 2.47a                                  | 26.04 ± 1.15c                                 |
|                   | 20–30           | 1.37 ± 0.03b                          | 21.50 ± 1.01c                        | 2.84 ± 0.53b                            | 22.50 ± 1.50a                           | 52.33 ± 4.70a                                  | 25.16 ± 5.31a                                 |
| Mid-alti-<br>tude | 0–10            | 1.50 ± 0.04a                          | 36.37 ± 2.62a                        | 5.33 ± 0.06a                            | 6.39 ± 3.19b                            | 29.89 ± 15.4b                                  | 63.72 ± 18.68a                                |
|                   | 10–20           | 1.58 ± 0.08a                          | 30.69 ± 0.33a                        | 4.42 ± 0.10a                            | 8.57 ± 1.47b                            | 35.90 ± 7.01b                                  | 55.52 ± 8.48a                                 |
|                   | 20–30           | 1.65 ± 0.07a                          | 29.51 ± 0.15a                        | 4.63 ± 0.33a                            | 10.53 ± 4.65b                           | 39.47 ± 16.71a                                 | 49.99 ± 21.37a                                |
| Low alti-<br>tude | 0–10            | 1.13 ± 0.07b                          | 22.03 ± 1.96b                        | 4.19 ± 0.03b                            | 12.8 ± 3.56a                            | 46.48 ± 1.09ab                                 | 40.71 ± 2.63b                                 |
|                   | 10–20           | 1.24 ± 0.07b                          | 24.62 ± 0.04b                        | 4.48 ± 0.57a                            | 11.27 ± 0.73b                           | 46.93 ± 5.56a                                  | 41.78 ± 6.29b                                 |
|                   | 20–30           | 1.25 ± 0.01b                          | 25.32 ± 0.72b                        | 3.48 ± 0.59ab                           | 9.17 ± 1.67b                            | 46.61 ± 6.58a                                  | 44.22 ± 6.72a                                 |
| Plot              | Soil depth (cm) | Total porosity (%)                    | Capillary porosity (%)               | Organic matter<br>(g•kg <sup>-1</sup> ) | Total Nitrogen<br>(g•kg <sup>-1</sup> ) | Total Phos-<br>phorus<br>(g•kg <sup>-1</sup> ) | Total Potas-<br>sium<br>(g•kg <sup>-1</sup> ) |
| High altitude     | 0–10            | 38.08 ± 7.69b                         | 24.36 ± 1.08b                        | 24.70 ± 1.83b                           | 0.54 ± 0.13b                            | 1.84 ± 0.07b                                   | 3.41 ± 0.43b                                  |
|                   | 10–20           | 36.52 ± 3.14b                         | 24.02 ± 2.94c                        | 23.00 ± 0.76a                           | 0.34 ± 0.20b                            | 1.63 ± 0.24b                                   | 2.68 ± 1.33b                                  |
|                   | 20–30           | 35.28 ± 6.28c                         | 22.22 ± 6.23c                        | 21.66 ± 0.85a                           | 0.27 ± 0.06b                            | 1.33 ± 0.33b                                   | 2.77 ± 0.90b                                  |
| Mid-alti-<br>tude | 0–10            | 63.88 ± 5.34ab                        | 53.26 ± 3.86a                        | 39.74 ± 7.52a                           | 3.44 ± 0.65a                            | 2.93 ± 0.49a                                   | 9.29 ± 0.21a                                  |
|                   | 10–20           | 58.54 ± 5.25a                         | 48.98 ± 3.86a                        | 43.20 ± 28.71a                          | 3.06 ± 0.93a                            | 2.82 ± 0.50a                                   | 9.49 ± 0.18a                                  |
|                   | 20–30           | 58.10 ± 0.13a                         | 49.67 ± 0.07a                        | 33.91 ± 21.36a                          | 1.80 ± 0.10a                            | 2.54 ± 0.41a                                   | 9.86 ± 0.18a                                  |
| Low alti-<br>tude | 0–10            | 53.60 ± 12.48a                        | 41.12 ± 6.76a                        | 8.94 ± 0.32c                            | 0.25 ± 0.01b                            | 0.96 ± 0.06c                                   | 2.16 ± 0.38c                                  |
|                   | 10–20           | 50.29 ± 3.59a                         | 38.78 ± 3.06b                        | 11.20 ± 2.76a                           | 0.17 ± 0.05b                            | 1.21 ± 0.11b                                   | 2.48 ± 0.35b                                  |
|                   | 20–30           | 46.50 ± 1.43b                         | 36.89 ± 3.26b                        | 14.59 ± 0.64a                           | 0.16 ± 0.24b                            | 1.04 ± 0.02b                                   | 2.75 ± 0.67b                                  |

Different letters indicate that the difference between different elevations of the same soil depth is significant at the 0.05 level. The same below

processed independently. Key root parameters included root length density and root weight density, which were measured using the WinRHIZO-Pro root scanning system. Soil bulk density, porosity, and saturated hydraulic conductivity were determined using the ring knife method. Soil organic matter content was measured via the potassium dichromate oxidation with external heating method, while concentrations of chemical elements were analyzed using an Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES)(Wang et al. 2015). The physicochemical properties of soils from each sample plot are summarized in Table 2.

### Data processing

Stained image processing was conducted using a combination of Adobe Photoshop 2024, Image-Pro Plus 6.0. Core operations included geometric correction, color replacement, and noise reduction. The processed images were converted into binary data matrices (0 and

255) with a resolution of 500×500 pixels. All preferential flow-related indicators were subsequently calculated from these matrices using Microsoft Excel. Root-related data were obtained using the WinRHIZO-Pro root scanning system(Pu et al. 2025). Final root morphological indicators, including root length density and other derived metrics, were calculated in Excel. Following the acquisition of stained image data and root system indicators, statistical analyses were performed using Excel and SPSS. Graphical visualizations and plots were generated using Origin software.

### Results

#### Characteristics of soil preferential flow

To comprehensively evaluate the influence of elevation gradients—specifically low, medium, and high altitudes—and their corresponding vegetation types



(shrubs, trees, and grasses) on the morphological and quantitative characteristics of soil preferential flow, field dye tracing experiments were conducted, and vertical profile images were obtained. A total of nine representative dye-stained profiles were selected for detailed analysis. Figure 2 presents the staining patterns of vertical profiles across the different elevation gradients. Overall, the development of soil preferential flow exhibited significant variation with elevation, following the trend: mid-elevation > low-elevation > high-elevation. Morphologically, in the low-elevation shrub area (D1–D3), preferential flow paths were densely distributed and displayed a dendritic pattern. Profile excavation revealed sparse plant roots but abundant soil fauna activity (e.g., earthworms), which likely contributed to the formation of preferential pathways in place of roots. In the mid-altitude arbor area (Z1–Z3), a clear hierarchical structure was observed, with dominant vertical channels associated with main tree roots, reflecting the deep-rooted nature of the vegetation. In the high-altitude herbaceous area (G1–G3), preferential flow was primarily expressed as matrix flow interspersed with discontinuous vertical finger flow. Although preferential flow was present, it was less developed compared to the other elevations. The average maximum infiltration depths were 22.67 cm in low-altitude shrub areas (D1–D3), 34.93 cm in mid-altitude tree areas (Z1–Z3), and 10.17 cm in high-altitude herbaceous areas (G1–G3). These values were strongly correlated with soil porosity and mechanical composition. Compared to high-elevation plots, the dye-tracing profiles in mid- and low-elevation areas displayed greater spatial heterogeneity, deeper matrix flow zones, and enhanced dye penetration depth. This variation is likely attributable to the vertical zonation of climate induced by the altitudinal gradient in Mopan Mountain, which leads to distinct environmental conditions—particularly soil structure and biological activity—that significantly influence the formation and development of preferential flow pathways (Wang et al. 2023).

Figure 3 illustrates the trend in stained area ratio at consistent soil depths across sample plots located at different elevations. While variations in stained area ratio are observed among the different elevation sites at the same depth, there is an overall non-linear decreasing trend in stained area ratio with increasing soil depth—with the exception of high-altitude

sample plots (Lu et al. 2022). The “S”-shaped curves are especially evident in the low-elevation shrub plots (D1–D3) and mid-elevation tree plots (Z1–Z3). The clear upward rebound observed in these curves suggests possible lateral dye seepage (Guan et al. 2023). In contrast, high-elevation plots exhibit a smooth and rapid decrease in dye coverage with depth, indicating that preferential flow is minimal, and matrix flow dominates, thereby limiting the extent of dye infiltration.

Figure 4 presents a comparative analysis of several quantitative indicators of soil preferential flow derived from the dye-stained vertical profiles. Data are expressed as mean  $\pm$  standard deviation, with statistical significance denoted at  $p < 0.05$ . The stained area ratio quantifies the extent of dye penetration into each soil layer and is positively correlated with the degree of preferential flow. Larger values indicate a higher likelihood and intensity of preferential flow. A significant difference is evident between elevation zones. The average stained area ratio in high-altitude regions is 10.01%. Compared to 37.98% in mid-altitude regions, this difference is extremely significant ( $p < 0.001$ ). The preferential flow ratio reflects the proportion of the stained area attributed specifically to preferential pathways. It shows a similar trend to the stained area ratio, with 32.15% at high elevation, 48.84% at mid elevation, and again indicating a highly significant difference in the development of preferential flow across elevations.

#### Relationship between root systems and preferential flow

As illustrated in Fig. 5, root system indicators for trees, shrubs, and herbaceous plants vary significantly across different elevation gradients, with all indicators exhibiting a consistent decreasing trend with increasing soil depth. Root weight density is highest in high-altitude herbaceous vegetation, averaging  $7.5 \text{ mg}\cdot\text{cm}^{-3}$ , followed by mid-altitude trees at  $4.5 \text{ mg}\cdot\text{cm}^{-3}$ , and lowest in low-altitude shrubs at  $3.8 \text{ mg}\cdot\text{cm}^{-3}$ . Root volume density peaks in mid-altitude trees, averaging  $4.2 \text{ mm}^3\cdot\text{cm}^{-3}$ , followed by high-altitude herbs at  $0.8 \text{ mm}^3\cdot\text{cm}^{-3}$ , and is lowest in low-altitude shrubs at  $0.4 \text{ mm}^3\cdot\text{cm}^{-3}$ . Root length density and root surface area density are highest in high-altitude grasses, averaging  $1.4 \text{ cm}\cdot\text{cm}^{-3}$  and  $0.12$

$\text{cm}^2\cdot\text{cm}^{-3}$ , respectively. These values are followed by low-altitude shrubs ( $0.9$  and  $0.07 \text{ cm}^2\cdot\text{cm}^{-3}$ ), and are lowest in mid-altitude trees, at approximately  $0.06 \text{ cm}^2\cdot\text{cm}^{-3}$  and  $0.05 \text{ cm}^2\cdot\text{cm}^{-3}$ , respectively. These trends reflect the pronounced influence of elevation gradients on vertical root distribution, which in turn is closely linked to vegetation type. The observed patterns suggest distinct root resource allocation strategies that plants adopt to adapt to their respective environmental conditions.

Redundancy Analysis (RDA) was conducted to examine the relationships between root system characteristics and preferential flow parameters. As shown in Fig. 6, the RDA ordination diagram—supported by quantitative data in Table 3—reveals the regulatory roles of root traits on preferential flow. RDA1 and RDA2 explain 82.94% and 2.69% of the total variance, respectively. In the RDA biplot, root volume density exhibits a strong positive correlation with both preferential flow ratio and stained area ratio, while root weight density aligns closely with maximum infiltration depth. In the ordination space: Mid-elevation tree samples (Class 2) cluster clearly along the positive axis of RDA1. Low-elevation shrubs (Class 1) are positioned on the negative axis of RDA1, while high-elevation herbs (Class 3) are scattered along the negative axis of RDA2. These findings demonstrate a synergistic relationship between root volume density and key preferential flow parameters, providing strong evidence for a spatial coupling between root system architecture and the biotic mechanisms that govern preferential flow development.

#### Relationship between soil structure and preferential flow

To further investigate the effects of altitude gradients on soil physicochemical properties and their influence on preferential flow, we conducted a multivariate analysis based on experimental data from several soil indicators, including organic matter content, bulk density, total phosphorus, and multiple preferential flow parameters. The analysis was performed using Principal Component Analysis (PCA), as illustrated in Fig. 7. The results indicate that the first principal component (PC1) explains 82.2% of the total variance, the second principal component (PC2) accounts for 12.9%, together capturing a cumulative variance of 95.1%.

In the PCA biplot, high-altitude sample points are tightly clustered along a specific segment of the PC1 axis, while mid- and low-altitude points are distributed in distinct and separate areas. These distribution patterns clearly delineate the clustering range of samples according to elevation level. The directional arrows in the PCA plot represent the loading vectors for each soil indicator, illustrating both their directionality and relative contribution in the principal component space. These vectors also reveal the strength and orientation of the relationships between each soil property and elevation gradient. The analysis indicates that soil physicochemical properties differ markedly across elevation zones. Variables such as organic matter content and clay content appear to have stronger associations with high-elevation soils.

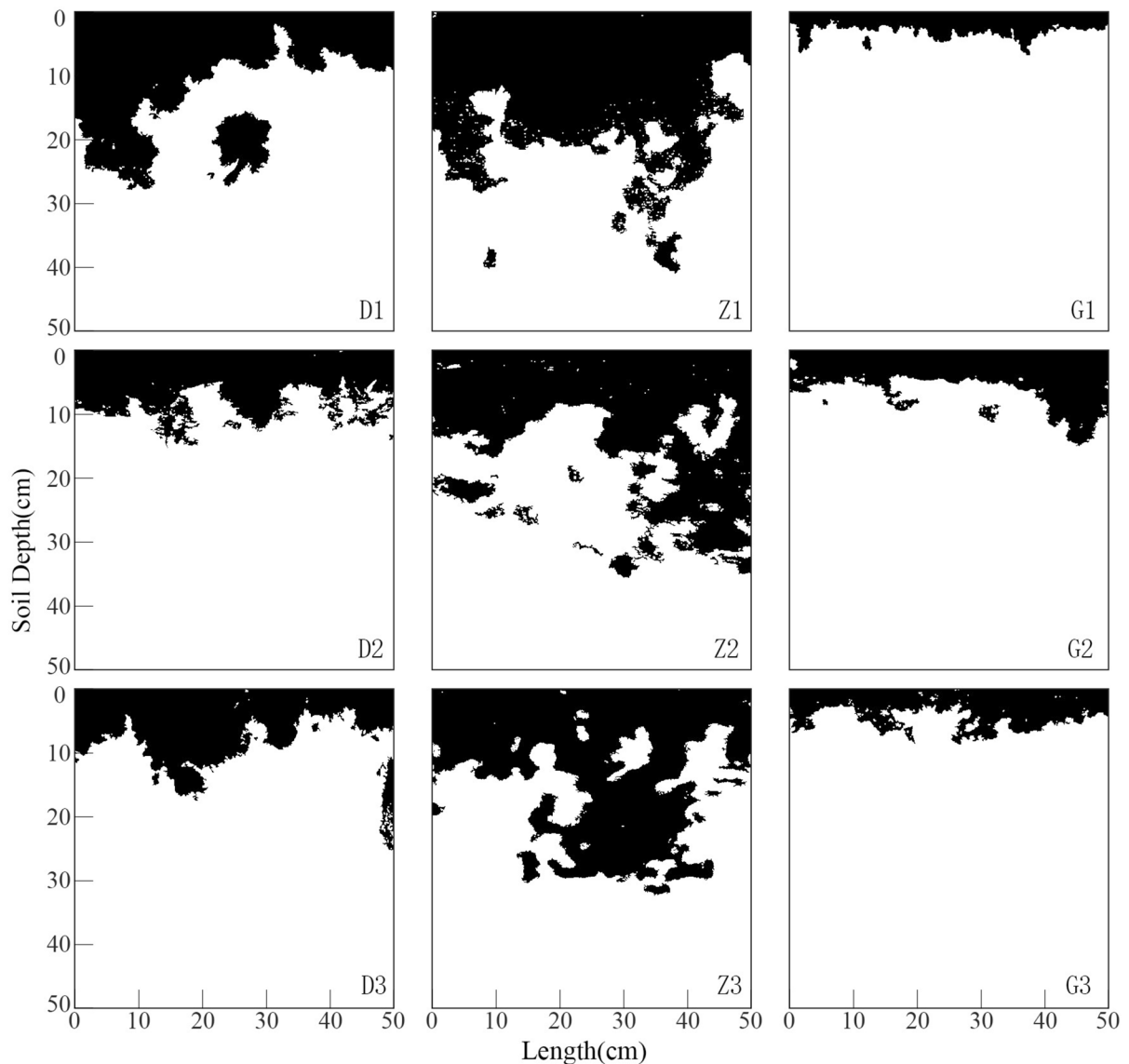
#### Impact of altitudinal gradient on preferential flow

To elucidate the influence of environmental conditions, soil physicochemical properties, and root system indicators on preferential flow across different altitudinal zones (high, mid, and low), Spearman correlation coefficient matrices were constructed for each elevation level, as shown in Fig. 8. The results reveal the following patterns:

In high-altitude herbaceous zones, the development of root systems significantly enhances preferential flow. Both root length density and root weight density exhibit strong positive correlations with the stained area ratio ( $p < 0.001$ ), suggesting that fine herbaceous roots form shallow, porous networks that facilitate water movement. However, the presence of clay appears to inhibit pore connectivity, likely due to pore blockage by fine particles in the rhizosphere. Organic matter indirectly promotes preferential flow by supporting root development: it is significantly correlated with root length density ( $p < 0.05$ ) and shows a positive—though not statistically significant—trend with stained area.

In mid-altitude environments, the deep root systems of trees play a critical role in improving soil structure. Root length density correlates positively with total porosity, which in turn exhibits a significant positive relationship with preferential flow ( $p < 0.05$ ). A synergistic relationship emerges among organic matter, root traits, and preferential flow parameters: Organic matter positively correlates with total porosity; Root weight density significantly correlates with the stained area ( $p < 0.05$ ); Organic



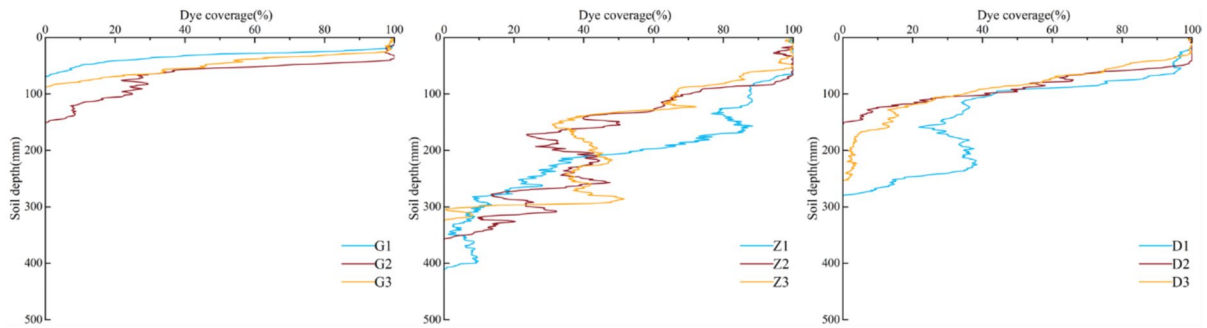


**Fig. 2** Vertical profile staining characteristics of soil preferential flow at different elevation gradients. D: low altitude; z: middle altitude; g: high altitude; Arabic numerals represent repetition. The same below

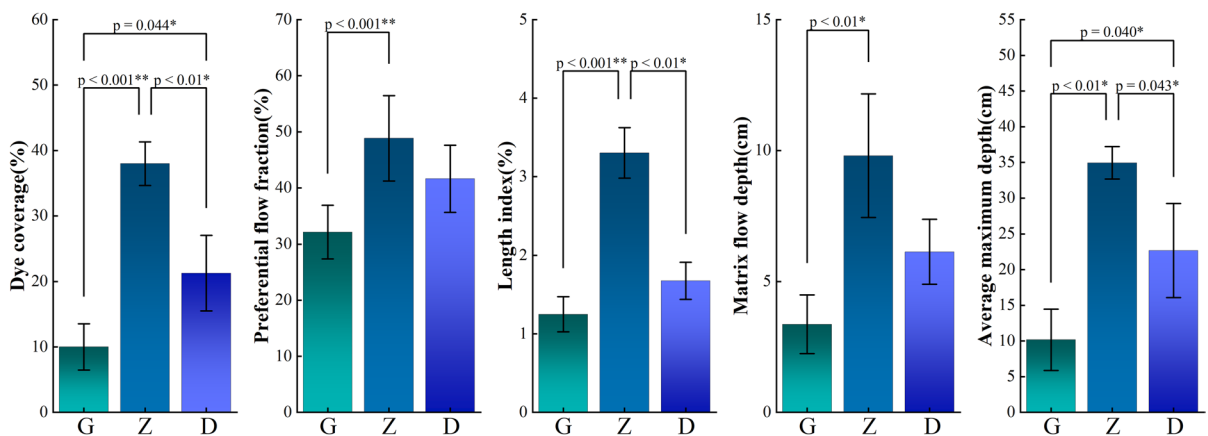
matter also correlates positively with aggregate stability ( $p < 0.05$ ), confirming that tree root systems enhance soil macroporosity and structural integrity, thereby promoting preferential flow. In low-altitude environments, unlike the patterns observed at higher elevations, excessive organic matter in low-elevation shrublands appears to inhibit root extension. A significant negative correlation is observed between organic matter and root length density ( $p < 0.05$ ), accompanied by reductions in root weight density and

other root-related parameters. Mechanistically, shrub litter with high lignin content decomposes slowly, creating physical barriers that block pore spaces. Nonetheless, the contribution of roots to preferential flow peaks in this zone, with both root length density and root weight density showing extremely strong positive correlations with the stained area ( $p < 0.001$ ).

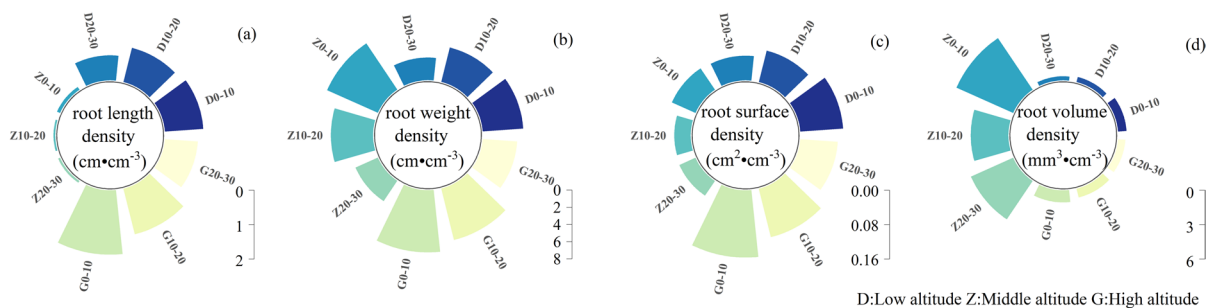
The altitudinal gradient regulates a cascading “root-soil structure-preferential flow” interaction across vegetation types. The altitude gradient



**Fig. 3** Variation in stained area ratio with depth across sample plots at different elevations



**Fig. 4** Parameters of soil preferential flow under different elevation gradients



**Fig. 5** Root system indicators across different elevation gradients. In the figure, 0–10, 10–20, 20–30 represent soil depths; (a)–(d) represent comparisons of four metrics for root systems at different elevations. Same as below

influences the cascade effect of “root-soil structure-preferential flow” across different vegetation types. In high-altitude herbaceous zones, root systems are the primary drivers of preferential flow, though this

is constrained by the high clay content in the soil. In mid-altitude forests, the deep root systems of trees generate macropore networks that enhance water flow connectivity, with porosity identified as the dominant

**Table 3** Explanatory power (%) of root traits on preferential flow parameters

| Variable | Explanation rate (%) | F     | P     |
|----------|----------------------|-------|-------|
| GTMD     | 82.1                 | 32.2  | 0.006 |
| GCMD     | 2.5                  | 1     | 0.398 |
| GZMD     | 1.1                  | 0.4   | 0.654 |
| GBMD     | 0.2                  | < 0.1 | 0.954 |

driver. However, a negative correlation with potassium may reduce root efficiency. In low altitude shrublands, high organic matter content restricts preferential flow through two pathways: inhibiting root growth and clogging soil pores directly. Despite these constraints, the positive correlations between root traits and preferential flow in low-elevation plots remain consistent with patterns observed at higher elevations.

## Discussion

### Regulatory mechanism of elevation gradient on preferential flow patterns

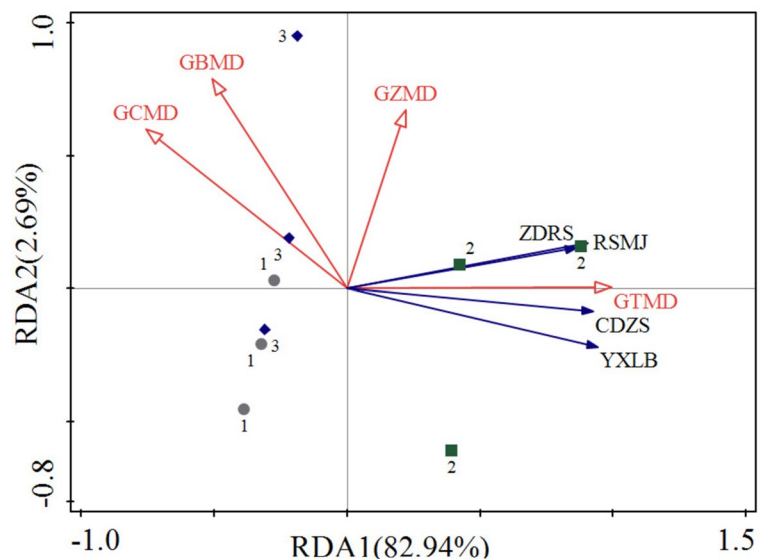
This study presents, for the first time, clear evidence of the significant regulatory role that elevation gradients play in shaping soil preferential flow patterns in the Mopan Mountain region of central Yunnan. The

development of preferential flow was found to be most prominent in mid-elevation forests, where the preferential flow ratio reached 48.84% and the dye-stained area ratio was 37.98%. This was followed by low-elevation shrublands, while high-elevation meadows exhibited the weakest preferential flow responses. This spatial heterogeneity can be attributed to the combined effects of the following three key factors:

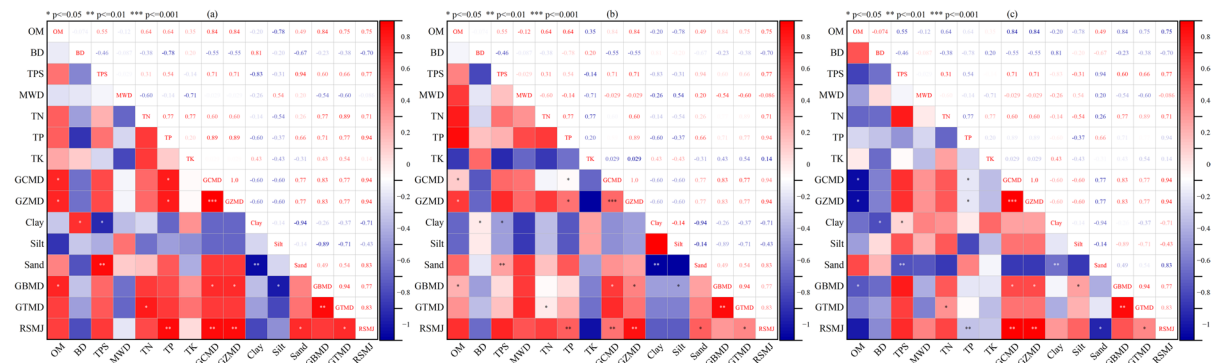
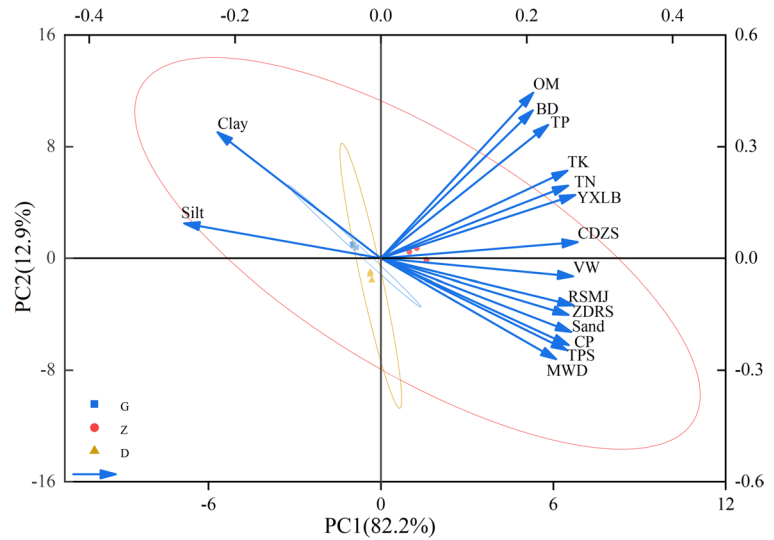
At mid elevations, the deep root systems of trees—with an average root volume density of  $4.2 \text{ mm}^3 \cdot \text{cm}^{-3}$ —form multi-tiered macropore networks that significantly improve soil connectivity for water flow. This finding is consistent with the results of Zhang et al., who reported that “for every  $1 \text{ mm}^3 \cdot \text{cm}^{-3}$  increase in root volume density, the number of preferential flow paths increases by 23%” in temperate forests (Zhang et al. 2015). Moreover, RDA in the current study confirms that root volume density alone accounts for 82.1% of the variation in preferential flow, further validating the conclusions of Shi et al., who found that the complex root systems in natural secondary karst forests increased the preferential flow ratio to 72.33% (Shi et al. 2024).

Soil structure differentiation: The mid-elevation soils exhibit a total porosity of 58.54%, with over 40% comprised of non-capillary (macropore) pores, creating favourable conditions for rapid preferential water movement. By contrast, although plant roots at high elevations are relatively abundant, they tend to be finer and more superficial, limiting their effectiveness in establishing macropore flow channels. Additionally, clay content in high-elevation soils

**Fig. 6** Redundancy analysis (RDA) biplot showing the relationship between root traits and preferential flow parameters. In the figure: GCMD: root length density; GBMD: root surface area density; YXLB: preferential flow ratio; CDZS: preferential flow length index; RSMJ: stained area ratio; ZDRS: maximum depth of infiltration. GZMD: root weight density; GTMD: root bulk density. YXLB: preferential flow ratio; CDZS: preferential flow length index; RSMJ: stained area ratio; ZDRS: maximum depth of infiltration. The same below



**Fig. 7** Principal component analysis of soil physicochemical properties and preferential flow indicators. OM: organic matter; BD: soil bulk density; TP: total phosphorus; TK: total potassium; TN: total nitrogen; YXLB: Priority flow ratio; CDZS: length index; RSMJ: dyeing area ratio; MWD: water-stable aggregates; TPS: Total porosity; CP: capillary porosity; VW: Volumetric water content; Sand: sand; Clay: clay; Silt: powder. The same below



**Fig. 8** Spearman correlation matrices between root, soil, and preferential flow indicators across different elevation gradients. (a) is high elevation, (b) is medium elevation, (c) is low elevation

reaches 18.13%, and bulk density is higher than in lower elevations, leading to compaction and reduced infiltration capacity. These findings align with those of Chen et al. who observed that “valley-bottom clay soils in tropical mountainous areas exhibit restricted preferential flow due to high compaction” (Chen et al. 2021).

**Organic matter-pore interaction:** Although low-elevation shrublands display a dense dendritic flow pattern, the preferential flow ratio remains lower (38.9%) than in mid-elevation forests. In the 20–30 cm soil layer, organic matter content reaches 14.59 g·kg<sup>-1</sup>, which improves nutrient retention; however, lignin degradation products from slow-decomposing shrub litter can clog macropores, impeding water

tion; red indicates positive correlation, blue indicates negative correlation, and shade indicates intensity

connectivity. This phenomenon supports the mechanism proposed by Bundt et al., who noted that “enrichment of organic matter reduces the hydraulic conductivity of macropores” (Bundt et al. 2001). Additionally, the S-shaped dye curves observed in the shrubland plots suggest the presence of lateral water movement, implying that a portion of the infiltrating dye may be diverted horizontally, thus reducing vertical penetration and weakening the overall preferential flow.

**Soil structure differentiation drives spatial heterogeneity in preferential flow**

By integrating the influences of root systems and soil physical properties, it becomes evident that soil

structure differentiation induced by altitudinal gradients serves as the primary driver of spatial heterogeneity in preferential flow. Therefore, it is essential to expand on the second mechanism previously outlined—soil physical structure—as a dominant control factor:

#### *Advantage of mid-altitude porosity*

The structural superiority of mid-altitude soils is strongly supported by PCA, which reveals high loading coefficients for total porosity (0.89) and non-capillary porosity (0.91) on the preferential flow ratio. This underscores the critical role of macropore connectivity in facilitating rapid and effective water infiltration—a conclusion consistent with findings by Lipiec et al. (Lipiec and Stepniewski 1995). Similarly, Lu et al., in a study conducted in the same Mopan Mountain region, reported that a 10% increase in soil porosity in broad-leaved forests led to a 37% increase in saturated hydraulic conductivity (Ks) (Lu et al. 2022). These findings align with our current results, which show that preferential flow is most pronounced at mid altitudes, where the preferential flow ratio reached 48.84%.

#### *Organic matter-induced clogging at low altitudes*

In low-altitude shrublands, the previously discussed clogging mechanism is further substantiated by the work of Bundt et al., who employed  $\delta^{13}\text{C}$  carbon isotope tracing to demonstrate that slow-decomposing, lignin-rich litter creates physical barriers that block macropore networks (Bundt et al. 2001). Additionally, they found that in soils with high organic matter content, the accumulation of particulate organic carbon (POC) within pores can reduce the hydraulic conductivity of preferential pathways by up to 42%.

Clay-induced limitation at high altitudes: High-altitude meadows contain significantly higher clay content compared to lower elevations, leading to increased soil compaction and bulk density (often exceeding  $1.27 \text{ g}\cdot\text{cm}^{-3}$ ). This results in a shallow maximum infiltration depth of only 10.17 cm. According to Lipiec et al., pore clogging by clay particles substantially increases resistance to water movement—doubling pore filling rates and fragmenting the continuity of preferential flow channels (Lipiec and Stepniewski 1995). This mechanism is consistent

with observations by Laine et al. in glacial till soils (Laine-Kaulio et al. 2015), and with findings by Chen et al., who noted that when clay content exceeds 15% in tropical mountain soils (Chen et al. 2021), the number of preferential flow paths decreases by 50%. These collective findings reinforce the conclusion that mechanical composition, particularly clay content, is a critical factor controlling preferential flow formation in vertically zoned mountainous ecosystems.

#### *Response mechanism of preferential flow patterns to altitude gradient*

Quantitative analysis using dye tracing has revealed that altitudinal gradients exert a substantial influence on the morphology of preferential flow, a pattern jointly regulated by root system architecture and soil pore structure. In the mid-elevation forest zone, preferential flow exhibits a distinct hierarchical structure, consisting of a dominant main macropore channel—with an average width exceeding 5 mm—accompanied by multi-level branching pathways. This configuration results in a stained area ratio of up to 37.98% and a maximum infiltration depth of 34.93 cm. These findings corroborate the conclusion by Shi et al., who reported in karst forests that “pipe flow created by deep tree roots is the core pathway for rapid water infiltration” (Shi et al. 2024). The RDA in this study further confirms that root volume density accounts for 82.1% of the variation in preferential flow, underscoring the pivotal role of deep-rooted trees in enhancing subsurface water connectivity. In contrast, in the high-altitude herbaceous zone, preferential flow is characterized by discontinuous vertical finger-like pathways, each less than 2 mm in width, with a maximum infiltration depth of only 10.17 cm. These flow patterns often appear as isolated stained patches—resembling “islands”—on the vertical profiles. This aligns with findings by Laine et al., who demonstrated that clay particle obstruction significantly restricts water movement in soils dominated by shallow root systems (Laine-Kaulio et al. 2015). In this study, a clay content of 18.13% effectively fills inter-root pores, promoting finger flow rather than continuous channel formation. Furthermore, the root length density of fine herbaceous roots is measured at  $1.4 \text{ cm}\cdot\text{cm}^{-3}$ , and these roots primarily occupy micropores ( $<0.5 \text{ mm}$ ), limiting the development of continuous flow channels. Chen et al. observed a

similar mechanism in tropical montane meadows, where clay content exceeding 15% reduced the maximum preferential flow depth by 48% compared to control plots (Chen et al. 2021). In low-altitude shrubland plots (D1–D3), despite the formation of dense dendritic preferential pathways, the preferential flow ratio is slightly lower (38.9%) than in mid-elevation forests (48.84%). This reduction is largely attributed to pore blockage by organic matter and lateral flow losses, as previously discussed. Additionally, the higher gravel content in low-elevation soils may further disrupt the continuity of vertical water pathways. Correlation analysis reveals a negative relationship between organic matter content and preferential flow ratio, which supports the pore-filling hypothesis proposed by Bundt et al. (Bundt et al. 2001).

## Conclusion

This study demonstrates that soil preferential flow occurs across all three elevation zones of Mopan Mountain, with the most developed preferential flow observed at mid elevations, where evergreen broad-leaved forests dominate. In this zone, the stained area ratio reached 37.98%, and the preferential flow ratio was 48.84%. The low-elevation shrubland exhibited moderate preferential flow development, while the high-elevation meadow displayed the weakest. The enhanced preferential flow observed at mid elevation is primarily attributed to the multi-layered macropore networks formed by deep-rooted trees, which significantly improve soil pore connectivity. This is supported by a root volume density of  $4.2 \text{ mm}^3\text{cm}^{-3}$  and an RDA explanatory power of 82.1%, confirming the dominant role of root volume density in regulating preferential flow. In contrast, although high-altitude herbaceous vegetation showed elevated values in several root metrics, their shallow fine roots—concentrated in micropores  $< 0.5 \text{ mm}$ —with a root length density of  $1.4 \text{ cm}\text{cm}^{-3}$ , limit the formation of continuous preferential flow paths. These findings confirm that elevation gradient strongly influences the intensity and morphology of preferential flow, primarily by shaping vegetation structure and root architecture.

Soil physicochemical properties also play a critical role. At mid elevation, soils exhibit a bulk density of  $1.50 \text{ g}\text{cm}^{-3}$  and a total porosity of up to 58.54%, with a high proportion of non-capillary pores that facilitate

rapid infiltration. In contrast, low-elevation soils contain  $14.59 \text{ g}\text{kg}^{-1}$  of organic matter in the 20–30 cm layer, which contributes to pore blockage and inhibits preferential flow. High-altitude soils, with a clay content of 18.13% and bulk density exceeding  $1.27 \text{ g}\text{cm}^{-3}$ , are prone to compaction. The pore-filling effect in these soils significantly increases infiltration resistance, resulting in a maximum infiltration depth of only 10.17 cm. In summary, soil structure differentiation, driven by altitudinal gradients, defines the physical constraints of preferential flow, while root architecture, particularly root volume density, constitutes the primary biological driver.

**Author's contributions** All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by [Jindong Xiang], [Miaomiao Zhai] and [Zhenchao Wang]. Project administration, funding acquisition, supervision, and review by [Yangyi Zhao] and [Keqin Wang]; The first draft of the manuscript was written by [Jindong Xiang] and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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**Data availability** The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Competing interests** Authors declare that they have no competing interests. The authors have no relevant financial or non-financial interests to disclose.

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